1 Substitution

1. Consider the indefinite integral $\int e^{x^2}(2x)dx$. To evaluate the integral, we need to find a function $F(x)$ such that $F'(x) = f(x) = e^{x^2}(2x)$. Can you find such a function $F(x)$? (*Hint:* Use the Chain Rule.)

Theorem 1.1 (The Substitution Rule)**.** *Suppose that g is continuously differentiable on an interval I, f is continuous on the range of g, and F is an antiderivative for f on the range of g. Then*

$$\int f(g(x))g'(x)dx = F(g(x)) + C$$

2. Prove Theorem 1.1.

Here is how to remember Theorem 1.1: Suppose we have an integral of the form $\int f(g(x))g'(x)dx$ and that F is an antiderivative of f. If we let $u = g(x)$, then the differential of u is $du = g'(x)dx$. We therefore see that

$$\int f(\underbrace{g(x)}_{u})\underbrace{g'(x)dx}_{du} = \int f(u)du = F(u) + C = F(g(x)) + C.$$

Thus, we can think of the Substitution Rule as a change of variables from $x \mapsto u = g(x)$ if we treat the 'dx' in the integral as if it were a differential (as the notation suggests).

Problem-Solving Strategy: Integration by Substitution

(i) Look carefully at the integrand and select an expression $g(x)$ within the integrand to set equal to u. Select $g(x)$ such that $g'(x)$ is also part of the integrand.

(ii) Substitute $u = g(x)$ and $du = g'(x)dx$ into the integral.

(iii) We should now be able to evaluate the integral with respect to u. If the integral can't be evaluated, go back and select a different expression to use as u.

(iv) Evaluate the integral in terms of u.

(v) Substitute back $u = g(x)$ to write the final result in terms of x.

3. Use the Substitution Rule to evaluate each of the following indefinite integrals.

   (a) $\int 2x(x^2-3)^3 dx$

   (b) $\int 6x(3x^2+4)^4 dx$

Sometimes we need to adjust the constants in our integral if they don't match up exactly with the expressions we are substituting.

Example 1.2. Consider $\int (x^3+5)^9 x^2 dx$. If we let $u = x^3+5$, then $du = 3x^2 dx$. Notice that this substitution doesn't quite work, as original expression has only $x^2 dx$. Fortunately, this problem can be fixed simply by dividing both sides of $du = 3x^2$ by 3: $\frac{du}{3} = x^2 dx$. Thus

$$\int (\underbrace{x^3+5}_{u})^9 \underbrace{x^2 dx}_{\frac{du}{3}} = \frac{1}{2}\int u^9 du = \frac{1}{2}\cdot\frac{u^{10}}{10} + C = \frac{(x^3+5)^{10}}{20} + C.$$

4. Evaluate $\int z\sqrt{z^2-5}\,dz$.

5. Evaluate $\int \frac{\sin(t)}{\cos^3(t)} dt$

Sometimes we need to manipulate an integral in ways that are more complicated that just multiplying or dividing by a constant. We need to eliminate all the expressions within the integrand that are in terms of the original variable. When we are done, u should be the only variable in the integrand. In some cases, this means solving for the original variable in terms of u. This technique should become clear in the next example.

Example 1.3. Consider the integral $\int \frac{x}{\sqrt{x-1}}dx$. If we let $u = x-1$, then $du = dx$. This, however, does not account for the x in the numerator of the integrand. We need to express x in terms of u. Since $u = x-1$, this means that $x = u+1$. Now we can rewrite the integral in terms of u:

$$\begin{aligned}\int \frac{x}{\sqrt{x-1}}dx &= \int \frac{u+1}{\sqrt{u}}du = \int \left(\frac{u}{\sqrt{u}} + \frac{1}{\sqrt{u}}\right)du \\ &= \int \left(u^{1/2} + u^{-1/2}\right)du = \int u^{1/2}du + \int u^{-1/2}du \\ &= \frac{2u^{3/2}}{3} + 2u^{1/2} + C \\ &= \frac{2(x-1)^{3/2}}{3} + 2(x-1)^{1/2} + C.\end{aligned}$$

6. Evaluate $\int \sqrt{1+x^2}x^5dx$.

2 Substitution for Definite Integrals

Theorem 2.1 (The Substitution Rule for Definite Integrals)**.** *If g is continuously differentiable on $[a,b]$ and f is continuous on the range of $u = g(x)$ with antiderivative F, then*

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du = F(g(b)) - F(g(a)).$$

7. Prove Theorem 2.1.

Example 2.2. Consider the definite integral $\int_0^1 x^2(1+2x^3)^5 dx$. If we let $u = g(x) = 1+2x^3$, then $du = 6x^2dx$ and therefore $\frac{du}{6} = x^2dx$. Since $g(0) = 1$ and $g(1) = 1+2 = 3$, It follows that

$$\int_0^1 x^2(1+2x^3)^5 dx = \frac{1}{6}\int_1^3 u^5 du = \frac{1}{6}\cdot\frac{u^6}{6}\Bigg|_1^3 = \frac{1}{36}(3^6 - 1^6) = \frac{182}{9}.$$

8. Evaluate $\int_1^2 \frac{dx}{(3-5x)^2}$.

3 Symmetry

The next theorem uses the Substitution Rule to Definite Integrals (Theorem 2.1) to simplify the calculation of integrals of functions that possess symmetry properties.

Theorem 3.1 (Integrals of symmetric functions)**.** *Suppose f is continuous on $[-a,a]$.*

(a) If f is even, then $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$.

(b) If f is odd, then $\int_{-a}^{a} f(x)dx = 0$.

Theorem 3.1 is illustrated below. For the case where f is positive and even, part (a) says that the area under $y = f(x)$ from $-a$ to a is twice the area from 0 to a because of symmetry. For part (b), because the integral $\int_{a}^{b} f(x)dx$ can be expressed as the area above the x-axis and below $y = f(x)$ minus the area below the x-axis and above the graph, the total area is 0 since these cancel.

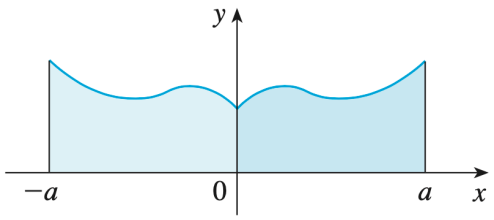


Figure 1: f even, $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx$

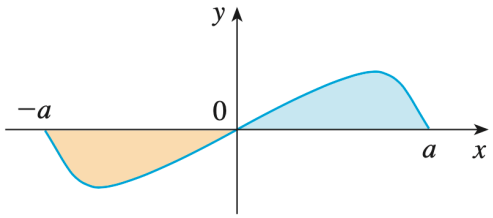


Figure 2: f odd, $\int_{-a}^{a} f(x)dx = 0$

Caution: Note that Theorem 3.1 only applies when the interval we are integrating over is of the form $[-a,a]$; i.e., it is symmetric about the origin. E.g., $f(x) = x$ is odd, but $\int_{-1}^{2} xdx = \left[\frac{x^2}{2}\right]_{-1}^{2} = \frac{4}{2} - \frac{1}{2} = \frac{3}{2} \neq 0$ since $[-1,2]$ is not symmetric about the origin.

9. Prove Theorem 3.1.

Example 3.2. Consider $\int_{-2}^{2}(x^6+1)dx$. Since the integrand $f(x)=x^6+1$ is even ($f(-x)=(-x)^6+1=x^6+1=f(x)$), by part (a) of Theorem 3.1 we have $\int_{-2}^{2}(x^6+1)dx=\int_{0}^{2}(x^6+1)dx=2\left[\frac{x^7}{7}+x\right]_0^2=2(\frac{128}{7}+2)=\frac{248}{7}$.

10. Evaluate $\displaystyle\int_{-1}^{1}\frac{\tan(x)}{1+x^2+x^4}dx$.